**CAPSTONE PROJECT PROPOSAL**

# Machine Learning Model for Refugee Population Forecasting and Humanitarian Resource Planning in Kenya

*Team: XG Boost Busters*

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Data Source: Kenya Refugee and Asylum Population Dataset (Humanitarian Data Exchange — HDX HAPI)

## 1. Business Understanding

**Problem / Question:** Kenya hosts one of the largest refugee populations in sub-Saharan Africa, yet humanitarian resource planning is typically based on current or historical population counts rather than forward-looking forecasts. This project asks: can a machine learning model accurately forecast Kenya's refugee and asylum-seeker population disaggregated by origin country, population group, gender, and age and can those forecasts be translated into concrete humanitarian resource-planning recommendations?

**Motivation:** Kenya continues to receive displaced people fleeing conflict, persecution, climate shocks, and political instability from neighboring countries such as Somalia, South Sudan, the Democratic Republic of Congo, Ethiopia, and Burundi. When resource planning lags behind population change, it directly limits the ability of humanitarian organizations to respond to sudden surges in arrivals. The team is motivated by the opportunity to apply data science to a real, high-stakes humanitarian problem improving preparedness for food, healthcare, shelter, and protection services rather than leaving planning reactive.

**Domain:** Humanitarian aid and refugee resettlement / international development analytics.

**Target Audience:** Government and humanitarian bodies responsible for refugee response in Kenya, including the Department of Refugee Services (DRS), the UNHCR, the World Food Programme (WFP), and the Kenya Red Cross Society (KRCS), along with policymakers and NGO planning teams.

**Real-World Impact:** If deployed, accurate forecasts would let humanitarian agencies pre-position food, healthcare, and shelter resources ahead of population surges instead of reacting after arrivals occur, shortening response times and improving the efficiency of resource allocation across demographic groups.

**Prior Work / Domain Knowledge:** The project draws on the UNHCR Age, Gender and Diversity (AGD) framework for disaggregated humanitarian planning, and builds on the structure of the HDX HAPI dataset itself, which is maintained specifically to support evidence-based humanitarian coordination. The team's exploratory analysis of population trends (2001–2025) and known regional displacement events, such as the South Sudanese civil war and the Tigray conflict, further informs feature design.

## 2. Data Understanding

**Data Source:** The Kenya Refugee and Asylum Population Dataset, obtained from the Humanitarian Data Exchange (HDX) HAPI platform, a public repository maintained to support humanitarian coordination and is already collected and accessible for download as a structured file.

**Coverage:** The raw dataset spans a 25-year period (2001–2025) and contains 27,664 records across 15 variables, with each record representing a population observation for a specific reporting period, location, population category, gender, and age group.

The dataset defines four population groups: Refugees (REF), Asylum Seekers (ASY), Host Community (HST), and Others of Concern (OOC).

### Key Features

| Column | Description |
|---|---|
| reference_period_start / end | Reporting period boundaries |
| asylum_location_code | Country where asylum is provided (filtered to Kenya) |
| origin_location_code | Country of origin of the displaced population |
| population_group | Refugee (REF), Asylum Seeker (ASY), Host Community (HST), or Other of Concern (OOC) |
| gender | Gender category (male / female) |
| age_range | Age cohort category (0–4, 5–11, 12–17, 18–59, 60+) |
| min_age / max_age | Age boundaries of the cohort |
| origin_has_hrp / origin_in_gho | Binary humanitarian indicators for the origin country |
| population (target) | Number of individuals recorded in the category |

**Prior Work on this Dataset:** As a maintained HDX HAPI product, the dataset underlies existing UNHCR situational reporting and dashboards, which are primarily descriptive. This project builds on that groundwork by moving from descriptive reporting to predictive forecasting, adding engineered temporal and demographic features and comparing traditional machine learning models against a deep-learning (FT-Transformer) approach.

## 3. Data Preparation

**Storage Format:** The data is stored as a structured, tabular file (CSV) with a mix of categorical, numerical, and date-based variables.

**Variable Types & Initial Statistics:** Categorical variables include origin location, population group, gender, and age range; numerical variables include the humanitarian indicators (origin_has_hrp, origin_in_gho) and the population target. Initial descriptive statistics show the population variable is highly right-skewed, ranging from 0 to 517,666 individuals, with a mean of roughly 1,410 against a median of 0 reflecting the highly disaggregated nature of the records.

**Anticipated Preprocessing Steps:**

- Filter records to asylum_location = Kenya so the analysis is scoped to the target country.
- Remove aggregated "all" categories in gender and age_range to avoid double-counting alongside individual-category records.
- Extract a year feature from the reporting period start date for temporal modeling.
- Convert binary humanitarian indicators (HRP/GHO status) from categorical/boolean to numeric form.
- Evaluate zero and extreme population values to decide whether they represent valid observations or data issues.
- One-hot encode categorical predictors for tree-based/linear models; label-encode and embed categorical predictors for the FT-Transformer.

**Anticipated Cleaning Challenges:** Roughly 44% of observations carry zero population counts, which are expected in a highly disaggregated dataset rather than missing data, but they still need to be retained thoughtfully so they don't distort model learning. The population target is also extremely right-skewed, with a small number of very large refugee populations (e.g., from Somalia and South Sudan) that must be preserved rather than treated as outliers, since they capture genuine large-scale displacement events that matter most for planning.

**Minimum Row Count:** After filtering to Kenya as the asylum location and removing aggregated "all" categories, an estimated 10000 usable rows are expected, threshold needed for a supervised regression project.

**Planned Visualization:** Univariate summaries (gender and origin-country distributions), bivariate breakdowns (top origin countries by population, population by group/gender/age), multivariate views (age × gender × population, origin × gender × population, and a treemap of origin-country concentration), a time-series view of population by year, and a correlation heatmap of the numerical features.

## 4. Modeling

**Problem Type:** This is a supervised regression problem; the target variable is continuous.

**Target Variable:** population: the recorded number of individuals within a given demographic/geographic category and reporting period.

**Baseline Model:** Linear Regression, run through a shared preprocessing pipeline (one-hot encoding for categorical, pass-through for numerical), will serve as the baseline for judging whether more complex models are worth the added complexity.

**Candidate Techniques:** Because the population target has a non-linear, right-skewed relationship with the demographic and temporal predictors, tree-based and ensemble methods (Decision Tree, Random Forest, XGBoost) are expected to outperform the linear baseline. A deep-learning approach, the FT-Transformer (Feature Tokenizer Transformer), will also be explored as a stretch goal to test whether learned feature embeddings and self-attention can outperform conventional tabular models.

| Model | Role | Status |
|---|---|---|
| Linear Regression | Baseline | Planned first |
| Decision Tree Regressor | Non-linear benchmark | Planned |
| Random Forest | Ensemble benchmark | Planned |
| XGBoost | Gradient-boosted benchmark | Planned |
| FT-Transformer | Deep-learning candidate for final model | Stretch goal |

## 5. Evaluation

**Success Metrics:** Model performance will be measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination ($R^2$), evaluated on a chronological train/test split (training on records through 2022, testing on records after 2022) so the evaluation mirrors a real forecasting scenario rather than random cross-validation.

**Minimum Viable Product (MVP):** Within the first week, the MVP is a cleaned and feature-engineered dataset, a working preprocessing pipeline, and a trained-and-evaluated Linear Regression baseline plus at least one tree-based model (e.g., Decision Tree or Random Forest), compared on MAE, RMSE, and $R^2$.

**Stretch / Level-Up Goals:**

- Train and tune XGBoost and compare against the tree-based ensemble models.
- Implement and tune the FT-Transformer deep-learning model as the top-performing candidate.
- Apply SHAP for model interpretability and to surface which demographic/humanitarian factors drive forecasts.
- Package the final model into an interactive Streamlit decision-support application.

## 6. Deployment

**Reporting Plan:** Final results will be reported through a structured project notebook (data understanding through evaluation), a non-technical presentation summarizing business insights for a humanitarian audience, and a project README documenting the deployment package.

**Deployment Plan:** The best-performing model will be deployed as a Streamlit web application, runnable locally or via Streamlit Community Cloud, so it can be evaluated as an interactive decision-support tool rather than a static notebook.

**Functionality:** Users will input demographic and humanitarian characteristics (origin country, population group, gender, age range, year, HRP/GHO status), and the application will return a predicted population estimate along with derived humanitarian resource-planning metrics to support evidence-based decision-making.

## 7. Tools / Methodologies

**Data Gathering, Cleaning & Exploration:** pandas, numpy, matplotlib, seaborn, and plotly for data wrangling, descriptive statistics, and visualization.

**Modeling:** scikit-learn (Linear Regression, Decision Tree, Random Forest, preprocessing pipelines, evaluation metrics), xgboost, and PyTorch for the FT-Transformer deep-learning model.

**Interpretability & Deployment:** shap for model explainability; joblib for saving preprocessing objects; Streamlit for the deployed application.

**Modeling Algorithms:** Linear Regression, Decision Tree Regression, Random Forest, XGBoost, and FT-Transformer (Transformer-based tabular deep learning).

**Compute Environment:** Analysis will be performed locally, with cloud-based GPU compute (e.g., Google Colab) used as needed to accelerate FT-Transformer training.

**Data Storage:** The dataset and project artifacts will be version-controlled and stored in a GitHub repository, with local copies retained for active development.